WORKED SOLUTION

STEP II 2023, Question 2

Pure • Trigonometric Substitution

STEP II 2023, Question 2

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Each right-hand side is a familiar multiple-angle formula in disguise: $\frac{2t}{1-t^2}$ is $\tan 2\alpha$ when $t = \tan\alpha$, $\frac{3t-t^3}{1-3t^2}$ is $\tan 3\alpha$, and $2t^2 - 1$ is $\cos 2\alpha$ when $t = \cos\alpha$. Substituting turns three coupled equations into a single trigonometric equation in one angle, which is easy to solve and, crucially, easy to *count*. The only care needed is to make sure the substitution captures every solution exactly once.

Solution

Part (i)

Every real number is $\tan\alpha$ for exactly one α in $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$, so we lose nothing by writing $x = \tan\alpha$ with α in that interval. For the first equation to make sense we need $x \neq \pm 1$, which says $\alpha \neq \pm\frac{1}{4}\pi$, so 2α is not an odd multiple of $\frac{1}{2}\pi$ and the double-angle formula applies:

$$y = \frac{2\tan\alpha}{1-\tan^2\alpha} = \tan 2\alpha.$$

Exactly the same step, applied to $y = \tan 2\alpha$ (with $y \neq \pm 1$ since z is defined), gives $z = \tan 4\alpha$, and then applied to z (with $z \neq \pm 1$) gives $x = \tan 8\alpha$. Since also $x = \tan\alpha$,

$$\tan\alpha = \tan 8\alpha,$$

as required. Two angles have the same tangent exactly when they differ by a multiple of π, so $8\alpha = \alpha + k\pi$ for some integer k, i.e.

$$\alpha = \frac{k\pi}{7}.$$

The condition $-\frac{1}{2}\pi < \alpha < \frac{1}{2}\pi$ means $-\frac{7}{2} < k < \frac{7}{2}$, so $k = -3, -2, \ldots, 3$.

Conversely, each of these seven values gives a genuine solution: $2\alpha = \frac{2k\pi}{7}$ and $4\alpha = \frac{4k\pi}{7}$ are never odd multiples of $\frac{1}{2}\pi$ (that would need $4k$ or $8k$ to equal 7 times an odd number,

so $7 \mid k$, forcing $k = 0$, which gives 0, not an odd multiple), so none of x, y, z equals ± 1, every denominator is non-zero, and the chain above runs backwards. Different k give different α in $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$, hence different x. So there are exactly seven solutions.

To quote them in the requested form, reduce 2α and 4α by multiples of π into $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$, using $\tan(\theta - \pi) = \tan\theta$. For example $\tan\frac{4\pi}{7} = \tan(-\frac{3\pi}{7})$ and $\tan\frac{8\pi}{7} = \tan\frac{\pi}{7}$. Writing $t_m = \tan\frac{m\pi}{7}$, the solutions (x, y, z) are:

k	(x, y, z)
0	$(0, 0, 0)$
1	$(t_1,\ t_2,\ t_{-3})$
2	$(t_2,\ t_{-3},\ t_1)$
3	$(t_3,\ t_{-1},\ t_{-2})$

k	(x, y, z)
-1	$(t_{-1},\ t_{-2},\ t_3)$
-2	$(t_{-2},\ t_3,\ t_{-1})$
-3	$(t_{-3},\ t_1,\ t_2)$

Notice that apart from $(0, 0, 0)$ they come as the cyclic rearrangements of (t_1, t_2, t_{-3}) and of (t_{-1}, t_{-2}, t_3), as they must, since the system is unchanged by $x \to y \to z \to x$.

Part (ii)

First the triple-angle formula: with $t = \tan\alpha$,

$$\tan 3\alpha = \frac{\tan 2\alpha + \tan\alpha}{1 - \tan 2\alpha \tan\alpha} = \frac{\frac{2t}{1-t^2} + t}{1 - \frac{2t^2}{1-t^2}} = \frac{2t + t - t^3}{1 - t^2 - 2t^2} = \frac{3t - t^3}{1 - 3t^2}.$$

As before, put $x = \tan\alpha$ with $-\frac{1}{2}\pi < \alpha < \frac{1}{2}\pi$. The equations require $1 - 3x^2$, $1 - 3y^2$, $1 - 3z^2$ to be non-zero; given that, the formula gives in turn $y = \tan 3\alpha$, $z = \tan 9\alpha$ and $x = \tan 27\alpha$. So $\tan\alpha = \tan 27\alpha$, giving $26\alpha = k\pi$, that is

$$\alpha = \frac{k\pi}{26}, \qquad -13 < k < 13.$$

That is 25 candidate values of α, all giving different values of x.

> **Ask yourself** Could some of these 25 candidates make a denominator vanish, so that they are not really solutions?

This is the point to check. A denominator $1 - 3\tan^2\beta$ vanishes when $\beta \equiv \pm\frac{1}{6}\pi \pmod{\pi}$, and the tangent itself is undefined when β is an odd multiple of $\frac{1}{2}\pi$. Here the angles involved are $\beta = 3^j \cdot \frac{k\pi}{26}$ for $j = 0, 1, 2$.

- $3^j k\pi/26 = \pm\frac{1}{6}\pi + m\pi$ would mean $3^{j+1}k = \pm 13 + 78m$. The left-hand side is a multiple of 3 but the right-hand side is not (as 78 is a multiple of 3 and 13 is not). Impossible.
- $3^j k\pi/26 = \frac{1}{2}(2m+1)\pi$ would mean $3^j k = 13(2m+1)$, so 13 divides k, forcing $k = 0$; but then the left-hand side is 0, which is not an odd multiple of 13. Impossible.

So every candidate is a genuine solution, and the system has exactly 25 real solutions. (The value $k = \pm 13$, which the interval excludes, corresponds to $\alpha = \pm\frac{1}{2}\pi$, where $\tan\alpha$ does not exist.)

Part (iii)(a)

Now $|x| \leq 1$, so we may write $x = \cos\alpha$ for exactly one $\alpha \in [0, \pi]$. The double-angle formula $\cos 2\alpha = 2\cos^2\alpha - 1$ gives $y = \cos 2\alpha$, then $z = \cos 4\alpha$ and $x = \cos 8\alpha$; these automatically satisfy $|y| \leq 1$ and $|z| \leq 1$. Hence the solutions with all three in $[-1, 1]$ correspond exactly to the $\alpha \in [0, \pi]$ with

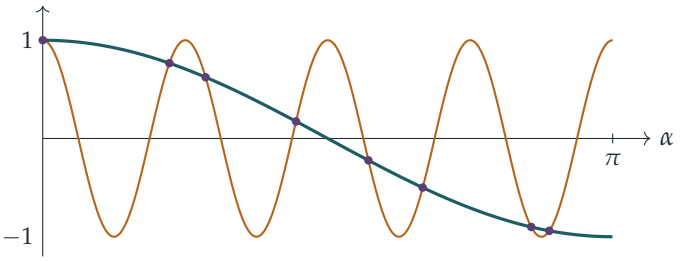
$$\cos 8\alpha = \cos\alpha.$$

Two angles have equal cosines exactly when they are equal or opposite up to a multiple of 2π, so

$$8\alpha = \alpha + 2m\pi \quad \text{or} \quad 8\alpha = -\alpha + 2n\pi, \qquad \text{i.e.} \qquad \alpha = \frac{2m\pi}{7} \quad \text{or} \quad \alpha = \frac{2n\pi}{9}.$$

In $[0, \pi]$ the first family gives $m = 0, 1, 2, 3$, that is $0, \frac{2\pi}{7}, \frac{4\pi}{7}, \frac{6\pi}{7}$, and the second gives $n = 0, 1, 2, 3, 4$, that is $0, \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{6\pi}{9}, \frac{8\pi}{9}$. The only overlap is $\alpha = 0$ (a common value would need $\frac{m}{7} = \frac{n}{9}$, i.e. $9m = 7n$, forcing $7 \mid m$). That leaves $4 + 5 - 1 = 8$ distinct values of α, and since cosine is one-to-one on $[0, \pi]$ they give 8 distinct values of x, each determining y and z. So there are exactly 8 solutions with $|x|, |y|, |z| \leq 1$.

The picture shows the eight crossings of $\cos 8\alpha$ (amber) with $\cos\alpha$ (teal) on $[0, \pi]$.



Part (iii)(b)

Now drop the restriction. Eliminating y and z: $y = 2x^2 - 1$ is a quadratic in x, so $z = 2y^2 - 1$ is a polynomial of degree 4 in x, and then $x = 2z^2 - 1$ becomes

$$x = 2\Big(2\big(2x^2 - 1\big)^2 - 1\Big)^2 - 1,$$

a polynomial equation of degree 8 in x (the leading term on the right is $2 \cdot (2 \cdot 2^2)^2 x^8 = 128x^8$, which nothing cancels). Expanded, it is

$$128x^8 - 256x^6 + 160x^4 - 32x^2 - x + 1 = 0.$$

Every solution (x, y, z) of the system has x a root of this octic, and x determines y and z. A polynomial of degree 8 has at most 8 real roots, and part (a) has already produced 8 distinct real roots, namely $\cos\alpha$ for the eight values of α found there. So these are *all* the roots, and every solution of the system is one of the eight from (a). In particular every solution has $|x| \leq 1$, $|y| \leq 1$, $|z| \leq 1$, which is what we had to show.

Ask yourself Why does (b) need the degree, rather than just the eight solutions from (a)?

Part (a) only found the solutions inside $[-1,1]$; on its own it says nothing about $x=5$, say. The degree bound is what rules out anything else: eight roots are all an octic can have, and (a) has used them up.

Checks

- In (i), $k=1$: $\tan\frac{\pi}{7}\approx 0.4816$, and $\frac{2(0.4816)}{1-0.4816^2}\approx 1.2540=\tan\frac{2\pi}{7}$. Two more steps return to 0.4816.
- A computer scan of the real line for roots of $f(f(f(x)))=x$ found exactly 7 roots for $f(t)=\frac{2t}{1-t^2}$ and exactly 25 for $f(t)=\frac{3t-t^3}{1-3t^2}$, the latter matching $\tan\frac{k\pi}{26}$, $|k|\le 12$.
- In (iii), $x=1$ gives $y=z=1$, the $\alpha=0$ solution, and indeed $128-256+160-32-1+1=0$. A scan of $[-50,50]$ for roots of the octic found exactly 8, all inside $[-1,1]$.

Common slips

- In (ii), allowing $k=\pm 13$ (or not restricting α at all), giving 27 solutions; $\alpha=\pm\frac{1}{2}\pi$ does not correspond to any real x.
- In (iii)(a), forgetting the second family $8\alpha=-\alpha+2n\pi$, or counting $\alpha=0$ twice.
- In (i), leaving answers such as $\tan\frac{4\pi}{7}$ unreduced, when the angle must lie in $(-\frac{1}{2}\pi,\frac{1}{2}\pi)$.

Beyond STEP

The polynomials T_n with $T_n(\cos\alpha)=\cos n\alpha$ are the *Chebyshev polynomials*; part (iii) is the equation $T_8(x)=x$, and the octic above is $T_8(x)-x$. Iterating a map like $x\mapsto 2x^2-1$ and asking for points that return after three steps is the starting point of the theory of dynamical systems: this map is "chaotic" on $[-1,1]$, and the angle substitution turns it into the simple doubling $\alpha\mapsto 2\alpha$. The same trick explains why the tangent maps in (i) and (ii) have so many periodic points. Siklos's *Advanced Problems in Mathematics* has more questions in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.